**C 23354** **(Pages : 2)** **Name.....**

**Reg. No.....**

**SECOND SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, APRIL 2022**

**April 2021 Session for SDE/Private Students**

(CBCSS)

Mathematics

MTH 2C 07—REAL ANALYSIS—II

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

**General Instructions**

**Covid Instructions are not applicable for Pvt/SDE students**

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section / Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section / sub part / sub division may be ignored.*
4. *There will be an overall ceiling for each Section / Part that is equivalent to the maximum weightage of the Section / Part.*

**Part A (Short Answer Questions)**

*Answer* **all** *questions.*
*Each question carries* 1 *weightage.*

1. Define Lebesgue outer measure and show that it is monotone.
2. Prove that translate of a measureable set is measureable.
3. State and prove a necessary condition for a function $f$ to be measureable.
4. Define Riemann integrability and show that the Dirichlet's function is not Riemann integrable.
5. Show by an example that strict inequality can occur in Fatou's Lemma.
6. When can we say that a family of measurable functions on E to be is tight over E. State the Vitali Convergence Theorem for sequence of functions defined on any set E.
7. Define functions of bounded variation and prove that monotone functions are of bounded variation.
8. State and Prove Riesz - Fischer Theorem.

(8 × 1 = 8 weightage)

**Turn over**

## Part B

*Answer any* **six** *question, choosing two questions from each unit.*
*Each question has weightage* 2.

UNIT I

9. Prove that Lebesgue outer measure is countably sub additive.

10. State and Prove the Outer approximation of a measureable set by Open sets and $G_\delta$ sets.

11. State and Prove Simple Approximation Lemma.

UNIT II

12. State and prove Monotone convergence Theorem.

13. Define Uniform integrability of a family of measureable functions on E and show that point wise limit of a sequence of uniformly integrable functions on E is integrable.

14. Prove that a bounded function defined on a set of finite measure E is Lebesgue integrable over E if and only if it is measurable.

UNIT III

15. Prove that a monotone function on $(a, b)$ is continuous except at a countable number of points in its domain.

16. Prove that a function is of bounded variation on a closed bounded interval $[a, b]$ if and only if it is the difference of two increasing functions on $[a, b]$.

17. State and prove Holder inequality and derive Cauchy Schwarz Inequality.

(6 × 2 = 12 weightage)

## Part C

*Answer any* **two** *from the following four questions (18-21).*
*Each question has weightage* 5.

18. Define a Sigma Algebra and prove the class of all Lebesgue measureable sets forms sigma algebra.

19. Define measurability of functions and prove that the sum and product of two measurable and finite valued functions on E are measurable.

20. Let $f$ and $g$ are non negative functions defined on E. Then

    a) State and prove linearity property of integration.

    b) State and prove Monotonicity property of integration.

    c) State and prove Additivity over domains of integration.

21. State and prove Vitali Covering Lemma.

(2 × 5 = 10 weightage)